

Variables & Equations

ProMo

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Contents

1	Variables	2
2	root	2
3	physical	5
4	macroscopic	7
5	reactions	12
6	macroscopic-reactions	13
7	macroscopic-control	14
8	Equations	15
9	Generic	15
10	Interface Link Equation	21

1 Variables

2 root

	var	symbol	documentation	type	units	eqs
10	$S_{I,q}$	S_Iq	selection matrix interface to control output	network		
17	cz_N	cz_N	output from control	network		
20	$A_{N,t,u}$	A_Ntu	mapping from input elements to outputs	network		
16	mv_I	mv_I	interface variable macro → control	network		
2	$F_{N,A}$	F	incidence matrix	network		
19	$A_{N,p,q}$	A_Npq	mapping from inputs to outputs	network		
11	$I_{t,u}$	I_tu	identity mapping from <t> to <u>	network		
4	$F^{sink}_{N,I}$	F_NI_sink	incidence matrix NI sink	network		
5	$F^{source}_{A,I}$	F_AI_source	incidence matrix AI source	network		
12	$S_{A,p}$	S_Ap	selection matrix interface species-related measures	network		
27	$I_{N,A}$	I_NA	identity mapping from <N> to <A>	network		
3	$F^{source}_{N,I}$	F_NI_source	incidence matrix NI source	network		

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	var	symbol	documentation	type	units	eqs
7	$F^{source}_{N,A}$	F_NA_source	incidence matrix NA source	network		
21	$u_{N,t,u}$	u_Ntu	input signal in control domain	network		
15	$S_{N,q,t}$	S_Nqt	selection matrix or splitter	network		
22	$y_{N,t,u}$	y_Ntu	output signal in control domain	network		
18	cz_I	cz_I	interface variable macro → control	network		
14	$S_{N,p,q}$	S_Npu	selection matrix for stacker	network		
6	$F^{sink}_{A,I}$	F_AI_sink	incidence matrix AI sink	network		
13	$S_{I,q}$	S_Aq	selection matrix arcs to outputs	network		
9	$S_{I,p}$	S_Ip	selection matrix interface to control input	network		
8	$F^{sink}_{N,A}$	F_NA_sink	incidence matrix NA sink	network		
1	t	t	time	frame	s	
107	Δt	t_interval	time interval	frame	s	6
105	t^o	to	starting time	frame	s	4
106	t^e	te	end time	frame	s	5

Continued on next page

	var	symbol	documentation	type	units	eqs
102	0	zero	numerical value zero	constant		1
101	#	value	numerical value	constant		
103	1	one	numerical value one	constant		2
104	0.5	oneHalf	numerical value one half	constant		3

3 physical

	var	symbol	documentation	type	units	eqs
24	r_{yN}	r_y	y-coordinate	frame	m	
23	r_{xN}	r_x	x-coordinate	frame	m	
25	r_{zN}	r_z	z-coordinate	frame	m	
137	m_N	m	mass	state	kg	30
109	S_N	S	fundamental state – internal entropy	state	$kg\,m^2\,K^{-1}\,s^{-2}$	
108	U_N	U	fundamental state – internal energy	state	$kg\,m^2\,s^{-2}$	
111	$n_{N,S}$	n	fundamental state – molar mass	state	mol	93
144	C_N	C	fundamental state – charge	state	$A\,s$	
110	V_N	V	volume	state	m^3	7
121	N^A	Avo	Avogadro constant	constant	mol^{-1}	
132	λ_S	Mm	molecular masses	constant	$kg\,mol^{-1}$	
122	k^B	Boltz	Boltzmann constant	constant	$kg\,m^2\,K^{-1}\,s^{-2}$	
123	R	R	gas constant	constant	$kg\,m^2\,mol^{-1}\,K^{-1}\,s^{-2}$	17

Continued on next page

	var	symbol	documentation	type	units	eqs
149	A_{xzN} A_{yzN} A_{xyN} ρ_N	Axz	cross sectional are xz	secondaryState	m^2	41
150		Ayz	cross sectional area yz	secondaryState	m^2	42
148		Axy	cross sectional area xy	secondaryState	m^2	40
143		rho	density	secondaryState	$kg\,m^{-3}$	36

4 macroscopic

	var	symbol	documentation	type	units	eqs
158	$c_{A,S}$	c_AS	concentration in convective event-dynamic flow	transport	$m^{-3} mol$	50
207	$\dot{H}_{zN}^d$	aHnd_z	accumulation of enthalpy due to diffusional mass flow in z-direction	transport	$kg m^2 s^{-3}$	102
157	d_A	d	flow direction of convective flow	transport		49
153	$\hat{q}_{zA}$	fq_z	heat flow in z-direction	transport	$kg m^2 s^{-3}$	45
204	$\dot{H}_{xN}^c$	aHnc_x	accumulation of enthalpy due to convective mass flow in x-direction	transport	$kg m^2 s^{-3}$	99
160	$\hat{n}_{xA,S}^c$	fnc_x	molar convective flow in x-direction	transport	$mol s^{-1}$	52
205	$\dot{H}_{xN}^d$	aHnd_x	accumulation of enthalpy due to diffusional mass flow in x-direction	transport	$kg m^2 s^{-3}$	100
211	$\hat{w}_A$	fw	a fixed work flow to start with	transport	$kg m^2 s^{-3}$	106
209	$\dot{q}_{yN}$	aq_y	accumulation due to heat flow in y-direction	transport	$kg m^2 s^{-3}$	104
214	$\dot{w}_N$	aw	accumulation of enthalpy due to work flow	transport	$kg m^2 s^{-3}$	109
213	$\dot{n}_{zN,S}^d$	and_z	accumulation due to diffusion in z-direction	transport	$mol s^{-1}$	108
156	$\hat{n}_{zA,S}^d$	fnd_z	diffusion flow in z-direction	transport	$mol s^{-1}$	48 91
151	$\hat{q}_{xA}$	fq_x	heat flow in x-direction	transport	$kg m^2 s^{-3}$	43

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	var	symbol	documentation	type	units	eqs
206	$\dot{H}_{yN}^d$	aHnd_y	accumulation of enthalpy due to diffusional mass flow in y-direction	transport	$kg\ m^2\ s^{-3}$	101
152	$\hat{q}_{yA}$	fq_y	heat flow in y-direction	transport	$kg\ m^2\ s^{-3}$	44
159	$\hat{V}_A$	fV	volumetric flow in x-direction	transport	$m^3\ s^{-1}$	51
194	$\dot{n}_{xN,S}^c$	anc_x	accumulation of molar mass due to convection	transport	$mol\ s^{-1}$	87
234	$\hat{m}_{N,A}$	fm	convective mass flow	transport	$kg\ s^{-1}$	133
210	$\dot{q}_{zN}$	aq_z	accumulation due to heat flow in z-direction	transport	$kg\ m^2\ s^{-3}$	105
195	$\dot{n}_{xN,S}^d$	and_x	accumulation due to diffusion in x-direction	transport	$mol\ s^{-1}$	88
212	$\dot{n}_{yN,S}^d$	and_y	accumulation due to diffusion in y-direction	transport	$mol\ s^{-1}$	107
155	$\hat{n}_{yA,S}^d$	fnd_y	diffusion flow in y-direction	transport	$mol\ s^{-1}$	47 90
154	$\hat{n}_{xA,S}^d$	fnd_x	diffusion flow in x-direction	transport	$mol\ s^{-1}$	46 89
208	$\dot{q}_{xN}$	aq_x	accumulation due to heat flow in x-direction	transport	$kg\ m^2\ s^{-3}$	103
181	$k_{yA,S}^d$	kdA_y	diffusivity in arc and y-direction	properties	$kg^{-1}\ m^{-4}\ mol^2\ s$	74
182	$k_{zA,S}^d$	kdA_z	diffusivity in arc and z-direction	properties	$kg^{-1}\ m^{-4}\ mol^2\ s$	75
187	k_{yA}^q	kqA_y	thermal conductivity in arc and y-direction	properties	$kg\ K^{-1}\ s^{-3}$	80

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	var	symbol	documentation	type	units	eqs
191	$\hat{k}_y^{d,Fick}{}_{A,S}$	kdAFick_y	Fick diffusivity in arc and y-direction	properties	ms^{-1}	84
192	$\hat{k}_z^{d,Fick}{}_{A,S}$	kdAFick_z	Fick diffusivity in arc and z-direction	properties	ms^{-1}	85
193	$h_{A,S}$	hA	partial molar enthalpies in arc	properties	$kg\,m^2\,mol^{-1}\,s^{-2}$	86
219	R_N^e	elResistant	electrical resistant	properties	$kg\,m^2\,A^{-2}\,s^{-3}$	115
185	k_{zA}^c	kcA_z	convective mass conductivity in arc and y-direction	properties	$m^{-1}\,s$	78
188	k_{zA}^q	kqA_z	thermal conductivity in arc and z-direction	properties	$kg\,K^{-1}\,s^{-3}$	81
183	k_{xA}^c	kcA_x	convective mass conductivity in arc and x diretion	properties	$m^{-1}\,s$	76
180	$k_{xA,S}^d$	kdA_x	diffusivity in arc and x-direction	properties	$kg^{-1}\,m^{-4}\,mol^2\,s$	73
189	ρ_A	rhoA	density in arc	properties	$kg\,m^{-3}$	82
190	$\hat{k}_x^{d,Fick}{}_{A,S}$	kdAFick_x	Fick's diffusivity in arc and x-direction	properties	ms^{-1}	83
184	k_{yA}^c	kcA_y	convective mass conductivity in arc and y-direction	properties	$m^{-1}\,s$	77
186	k_{xA}^q	kqA_x	thermal conductivity in arc and x-direction	properties	$kg\,K^{-1}\,s^{-3}$	79
115	H_N	H	Enthalpy	state	$kg\,m^2\,s^{-2}$	11 112
203	$n_{N,S}^o$	no	initial mass	state	mol	98

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	var	symbol	documentation	type	units	eqs
216	H^o_N	Ho	initial enthalpy	state	$kg\,m^2\,s^{-2}$	111
117	G_N	G	Gibbs free energy	state	$kg\,m^2\,s^{-2}$	13
116	A_N	A	Helmholtz energy	state	$kg\,m^2\,s^{-2}$	12
112	p_N	p	thermodynamic pressure	effort	$kg\,m^{-1}\,s^{-2}$	8
217	U^e_N	Ue	electrical potential – voltage	effort	$kg\,m^2\,A^{-1}\,s^{-3}$	113
113	T_N	T	temperature	effort	K	9 121
161	$\mu^o_{N,S}$	chemPotStandard	instantiating standard chemical potential	effort	$kg\,m^2\,mol^{-1}\,s^{-2}$	53
114	$\mu_{N,S}$	chemPot	chemical potential	effort	$kg\,m^2\,mol^{-1}\,s^{-2}$	10 54
124	C_{pN}	Cp	total heat capacity at constant pressure	secondaryState	$kg\,m^2\,K^{-1}\,s^{-2}$	18 117
138	$c_{N,S}$	c	molar concentration	secondaryState	$m^{-3}\,mol$	31
125	C_{VN}	CV	total heat capacity at constant volume	secondaryState	$kg\,m^2\,K^{-1}\,s^{-2}$	19
139	n^t_N	nt	total number of moles	secondaryState	mol	32
136	$h_{N,S}$	h	partial molar enthalpies	secondaryState	$kg\,m^2\,mol^{-1}\,s^{-2}$	29
222	T^{ref}_N	T_ref	reference temperature	secondaryState	K	119

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	var	symbol	documentation	type	units	eqs
118	v_{xN}	v_x	velocity in x-direction	secondaryState	ms^{-1}	14
142	c_{VN}	cV	specific heat capacity at constant volume	secondaryState	$m^2 K^{-1} s^{-2}$	35
119	v_{yN}	v_y	velocity in y-direction	secondaryState	ms^{-1}	15
140	$x_{N,S}$	x	mole fraction	secondaryState		33
120	v_{zN}	v_z	velocity in z-direction	secondaryState	ms^{-1}	16
141	c_{pN}	cp	specific heat capacity at constant pressure	secondaryState	$m^2 K^{-1} s^{-2}$	34 120
196	$\dot{n}_{N,S}$	an	differential mass balance without reaction	diffState	$mol s^{-1}$	92
220	$\dot{U}^e_A$	dUe	Kirkhoffs first law	diffState	$kg m^2 A^{-1} s^{-3}$	116
215	$\dot{H}_N$	dH	accumulation of enthalpy	diffState	$kg m^2 s^{-3}$	110
218	I^e_N	current	current definition	internalTransport	A	114
223	T^n_N	T_meas_norming	value to norm measurement of temperature	observation	K	122
224	$\bar{T}_N$	T_meas	temperature measurement	observation		123

5 reactions

	var	symbol	documentation	type	units	eqs
26	$N_{S,K}$	N	stoichiometric matrix	constant		60
198	K^o_K	Ko	Arrhenius frequency factor	constant	$m^{-3} mol s^{-1}$	
197	E^a_K	Ea	Arrhenius activation energy	constant	$kg m^2 mol^{-1} s^{-2}$	
167	$T_{N,p}$	T	link variable T to interface reactions	effort	K	
246	Ta	Ta	new	effort		

6 macroscopic-control

	var	symbol	documentation	type	units	eqs
244	<i>_mv_I_I</i>	<code>_mv_I</code>	link variable mv I to interface macroscopic »> control with source:node	get		143

7 macroscopic-reactions

	var	symbol	documentation	type	units	eqs
164	$_x$	$_x$	link variable x to interface macroscopic »> reactions with source:node	get		57
166	$_T$	$_T$	link variable T to interface macroscopic »> reactions with source:node	get	K	59
162	$_c$	$_c$	link variable c to interface macroscopic »> reactions with source:node	get	$m^{-3} mol$	55

8 Equations

9 Generic

no	equation	documentation	layer
1	$0 := \text{Instantiate}(\#, \#)$	numerical value zero	root
2	$1 := \text{Instantiate}(\#, \#)$	numerical value one	root
3	$0.5 := \text{Instantiate}(\#, \#)$	numerical value one half	root
4	$t^o := \text{Instantiate}(t, \#)$	starting time	root
5	$t^e := \text{Instantiate}(t, \#)$	end time	root
6	$\Delta t := \text{Instantiate}(t, \#)$	time interval	root
7	$V_N := r_{xN} \cdot r_{yN} \cdot r_{zN}$	volume	physical
8	$p_N := \frac{\partial U_N}{\partial V_N}$	thermodynamic pressure	physical
9	$T_N := \frac{\partial U_N}{\partial S_N}$	temperature	macroscopic
10	$\mu_{N,S} := \frac{\partial U_N}{\partial n_{N,S}}$	chemical potential	macroscopic
11	$H_N := U_N - p_N \cdot V_N$	Enthalpy	macroscopic
12	$A_N := U_N - T_N \cdot S_N$	Helmholtz energy	macroscopic
13	$G_N := U_N + p_N \cdot V_N - T_N \cdot S_N$	Gibbs free energy	macroscopic
14	$v_{xN} := \frac{\partial r_{xN}}{\partial t}$	velocity in x-direction	macroscopic
15	$v_{yN} := \frac{\partial r_{yN}}{\partial t}$	velocity in y-direction	macroscopic

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no	equation	documentation	layer
16	$v_{zN} := \frac{\partial r_{zN}}{\partial t}$	velocity in z-direction	macroscopic
17	$R := N^A \cdot k^B$	gas constant	physical
18	$C_{pN} := \frac{\partial H_N}{\partial T_N}$	total heat capacity at constant pressure	macroscopic
19	$C_{VN} := \frac{\partial U_N}{\partial T_N}$	total heat capacity at constant volume	macroscopic
29	$h_{N,S} := H_N \cdot (n_{N,S})^{-1}$	partial molar enthalpies	macroscopic
30	$m_N := \lambda_S \star^S n_{N,S}$	mass	macroscopic
31	$c_{N,S} := (V_N)^{-1} \cdot n_{N,S}$	molar concentration	macroscopic
32	$n_N^t := \text{reduceSum}(n_{N,S}, S)$	total number of moles	macroscopic
33	$x_{N,S} := (n_N^t)^{-1} \cdot n_{N,S}$	mole fraction	macroscopic
34	$c_{pN} := C_{pN} \cdot (m_N)^{-1}$	specific heat capacity at constant pressure	physical
35	$c_{VN} := C_{VN} \cdot (m_N)^{-1}$	specific heat capacity at constant volume	macroscopic
36	$\rho_N := (V_N)^{-1} \cdot m_N$	density	physical
40	$A_{xyN} := r_{xN} \cdot r_{yN}$	cross sectional area xy	physical
41	$A_{xzN} := r_{xN} \cdot r_{zN}$	cross sectional area xz	physical
42	$A_{yzN} := r_{yN} \cdot r_{zN}$	cross sectional area yz	physical
43	$\hat{q}_{xA} := k_{xA}^q \cdot A_{yzN} \cdot F_{N,A} \star^N T_N$	heat flow in x-direction	macroscopic
44	$\hat{q}_{yA} := k_{yA}^q \cdot A_{xzN} \cdot F_{N,A} \star^N T_N$	heat flow in y-direction	macroscopic

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no	equation	documentation	layer
45	$\hat{q}_{zA} := k_{zA}^q \cdot A_{xyN} \cdot F_{N,A} \star^N T_N$	heat flow in z-direction	macroscopic
46	$\hat{n}_{xA,S}^d := \hat{k}_x^{d,Fick}{}_{A,S} \cdot A_{yzN} \cdot F_{N,A} \star^N c_{N,S}$	Fick diffusion flow in x-direction	macroscopic
47	$\hat{n}_{yA,S}^d := \hat{k}_y^{d,Fick}{}_{A,S} \cdot A_{xzN} \cdot F_{N,A} \star^N c_{N,S}$	Fick diffusion flow in y-direction	macroscopic
48	$\hat{n}_{zA,S}^d := \hat{k}_z^{d,Fick}{}_{A,S} \cdot (A_{xyN} \cdot F_{N,A}) \star^N c_{N,S}$	Fick diffusion flow in z-direction	macroscopic
49	$d_A := \mathbf{sign} \left(F_{N,A} \star^N p_N \right)$	flow direction of convective flow	macroscopic
50	$c_{A,S} := (0.5 \cdot (F_{N,A} - d_A \cdot F_{N,A})) \star^N c_{N,S}$	concentration in convective event-dynamic flow	macroscopic
51	$\hat{V}_A := (\rho_A)^{-1} \cdot k_{xA}^c \cdot A_{yzN} \cdot F_{N,A} \star^N p_N$	volumetric flow in x-direction	macroscopic
52	$\hat{n}_{xA,S}^c := \hat{V}_A \cdot c_{A,S}$	molar convective flow in x-direction	macroscopic
53	$\mu_{N,S}^o := \mathbf{Instantiate}(\mu_{N,S}, \#)$	instantiating standard chemical potential	macroscopic
54	$\mu_{N,S} := \mu_{N,S}^o + R \cdot T_N \cdot \mathbf{ln}(x_{N,S})$	chemical potential standard model with mole fraction	macroscopic
73	$k_{xA,S}^d := I_{N,A} \star^N \left((\mu_{N,S})^{-1} \cdot \left(v_{xN} \cdot \left((V_N)^{-1} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \right) \right) \right)$	diffusivity in arc and x-direction	macroscopic
74	$k_{yA,S}^d := I_{N,A} \star^N \left((\mu_{N,S})^{-1} \cdot \left(v_{yN} \cdot \left((V_N)^{-1} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \right) \right) \right)$	diffusivity in arc and y-direction	macroscopic
75	$k_{zA,S}^d := I_{N,A} \star^N \left((\mu_{N,S})^{-1} \cdot \left(v_{zN} \cdot \left((V_N)^{-1} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \right) \right) \right)$	diffusivity in arc and z-direction	macroscopic
76	$k_{xA}^c := I_{N,A} \star^N \left(\left(\lambda_S \star^S (\mu_{N,S})^{-1} \right) \cdot (V_N)^{-1} \cdot \frac{\partial U_N}{\partial p_N} \cdot v_{xN} \right)$	convective mass conductivity in arc and x direction	macroscopic
77	$k_{yA}^c := I_{N,A} \star^N \left(\left(\lambda_S \star^S (\mu_{N,S})^{-1} \right) \cdot (V_N)^{-1} \cdot \frac{\partial U_N}{\partial p_N} \cdot v_{yN} \right)$	convective mass conductivity in arc and y-direction	macroscopic

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no	equation	documentation	layer
78	$k_{zA}^c := I_{N,A} \overset{N}{\star} \left(\left(\lambda_S \overset{S}{\star} (\mu_{N,S})^{-1} \right) \cdot (V_N)^{-1} \cdot \frac{\partial U_N}{\partial p_N} \cdot v_{zN} \right)$	convective mass conductivity in arc and y-direction	macroscopic
79	$k_{xA}^q := I_{N,A} \overset{N}{\star} \left((V_N)^{-1} \cdot C_{pN} \cdot v_{xN} \right)$	thermal conductivity in arc and x-direction	macroscopic
80	$k_{yA}^q := I_{N,A} \overset{N}{\star} \left((V_N)^{-1} \cdot C_{pN} \cdot v_{yN} \right)$	thermal conductivity in arc and y-direction	macroscopic
81	$k_{zA}^q := I_{N,A} \overset{N}{\star} \left((V_N)^{-1} \cdot C_{pN} \cdot v_{zN} \right)$	thermal conductivity in arc and z-direction	macroscopic
82	$\rho_A := I_{N,A} \overset{N}{\star} \rho_N$	density in arc	macroscopic
83	$\hat{k}_x^{d,Fick}{}_{A,S} := I_{N,A} \overset{N}{\star} \left(v_{xN} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \cdot (n_{N,S})^{-1} \right)$	Fick's diffusivity in arc and x-direction	macroscopic
84	$\hat{k}_y^{d,Fick}{}_{A,S} := I_{N,A} \overset{N}{\star} \left(v_{yN} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \cdot (n_{N,S})^{-1} \right)$	Fick diffusivity in arc and y-direction	macroscopic
85	$\hat{k}_z^{d,Fick}{}_{A,S} := I_{N,A} \overset{N}{\star} \left(v_{zN} \cdot \frac{\partial U_N}{\partial \mu_{N,S}} \cdot (n_{N,S})^{-1} \right)$	Fick diffusivity in arc and z-direction	macroscopic
86	$h_{A,S} := I_{N,A} \overset{N}{\star} h_{N,S}$	partial molar enthalpies in arc	macroscopic
87	$\dot{n}_{xN,S}^c := F_{N,A} \overset{A}{\star} \hat{n}_{xA,S}^c$	accumulation of molar mass due to convection	macroscopic
88	$\dot{n}_{xN,S}^d := F_{N,A} \overset{A}{\star} \hat{n}_{xA,S}^d$	accumulation due to diffusion in x-direction	macroscopic
89	$\hat{n}_{xA,S}^d := k_{xA,S}^d \cdot (A_{yzN} \cdot F_{N,A}) \overset{N}{\star} \mu_{N,S}$	Fick diffusion flow in x-direction	macroscopic
90	$\hat{n}_{yA,S}^d := k_{yA,S}^d \cdot (A_{yzN} \cdot F_{N,A}) \overset{N}{\star} \mu_{N,S}$	Fick diffusion flow in y-direction	macroscopic
91	$\hat{n}_{zA,S}^d := k_{zA,S}^d \cdot (A_{xyzN} \cdot F_{N,A}) \overset{N}{\star} \mu_{N,S}$	mass diffusion flow in z-direction	macroscopic
92	$\dot{n}_{N,S} := \dot{n}_{xN,S}^c + \dot{n}_{xN,S}^d$	differential mass balance without reaction	macroscopic

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no	equation	documentation	layer
93	$n_{N,S} := \int_{t^o}^{t^e} \dot{n}_{N,S} dt + n_{N,S}^o$	fundamental state – molar mass	macroscopic
98	$n_{N,S}^o := \text{Instantiate}(n_{N,S}, \#)$	initial mass	macroscopic
99	$\dot{H}_{xN}^c := F_{N,A} \overset{A}{\star} \left(\hat{n}_{xA,S}^S \overset{S}{\star} h_{N,S} \right)$	enthalpy accumulation due to convective flow in x-direction	macroscopic
100	$\dot{H}_{xN}^d := F_{N,A} \overset{A}{\star} \left(\hat{n}_{xA,S}^d \overset{S}{\star} h_{N,S} \right)$	accumulation of enthalpy due to diffusional mass flow in x-direction	macroscopic
101	$\dot{H}_{yN}^d := F_{N,A} \overset{A}{\star} \left(\hat{n}_{yA,S}^d \overset{S}{\star} h_{N,S} \right)$	accumulation of enthalpy due to diffusional mass flow in y-direction	macroscopic
102	$\dot{H}_{zN}^d := F_{N,A} \overset{A}{\star} \left(\hat{n}_{zA,S}^d \overset{S}{\star} h_{N,S} \right)$	accumulation of enthalpy due to diffusional mass flow in z-direction	macroscopic
103	$\dot{q}_{xN} := F_{N,A} \overset{A}{\star} \hat{q}_{xA}$	accumulation due to heat flow in x-direction	macroscopic
104	$\dot{q}_{yN} := F_{N,A} \overset{A}{\star} \hat{q}_{yA}$	accumulation due to heat flow in y-direction	macroscopic
105	$\dot{q}_{zN} := F_{N,A} \overset{A}{\star} \hat{q}_{zA}$	accumulation due to heat flow in z-direction	macroscopic
106	$\hat{w}_A := \text{Instantiate}(\hat{q}_{xA}, \#)$	a fixed work flow to start with	macroscopic
107	$\dot{n}_{yN,S}^d := F_{N,A} \overset{A}{\star} \hat{n}_{yA,S}^d$	accumulation due to diffusion in y-direction	macroscopic
108	$\dot{n}_{zN,S}^d := F_{N,A} \overset{A}{\star} \hat{n}_{zA,S}^d$	accumulation due to diffusion in z-direction	macroscopic
109	$\dot{w}_N := F_{N,A} \overset{A}{\star} \hat{w}_A$	accumulation of enthalpy due to work flow	macroscopic

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no	equation	documentation	layer
110	$\dot{H}_N := \dot{H}_{xN}^c + \dot{H}_{xN}^d + \dot{H}_{yN}^d + \dot{H}_{zN}^d + \dot{q}_{xN} + \dot{q}_{yN} + \dot{q}_{zN} + \dot{w}_N$	accumulation of enthalpy	macroscopic
111	$H_N^o := \mathbf{Instantiate}(H_N, \#)$	initial enthalpy	macroscopic
112	$H_N := \int_{t^o}^{t^e} \dot{H}_N dt + H_N^o$	Enthalpy	macroscopic
113	$U_N^e := (C_N)^{-1} \cdot U_N$	electrical potential – voltage	macroscopic
114	$I_N^e := \frac{dC_N}{dt}$	current definition	macroscopic
115	$R_N^e := (I_N^e)^{-1} \cdot U_N^e$	electrical resistant	macroscopic
116	$\dot{U}_N^e := F_{N,A} \overset{A}{\star} (R_N^e \cdot I_N^e)$	Kirkhoffs first law	macroscopic
117	$C_{pN} := m_N \cdot c_{pN}$	total heat capacity at constant pressure	macroscopic
119	$T_N^{ref} := \mathbf{Instantiate}(T_N, \#)$	reference temperature	macroscopic
120	$c_{pN} := \mathbf{Instantiate}(c_{pN}, \#)$	constant specific heat capacity at constant pressure	macroscopic
121	$T_N := H_N \cdot (C_{pN})^{-1} + T_N^{ref}$	temperature from constant heat capacity	macroscopic
122	$T_N^n := \mathbf{Instantiate}(T_N, \#)$	value to norm measurement of temperature	macroscopic
123	$\bar{T}_N := T_N \cdot (T_N^n)^{-1}$	temperature measurement	macroscopic
133	$\hat{m}_{N,A} := \hat{V}_A \cdot \rho_N$	convective mass flow	macroscopic

10 Interface Link Equation

no	equation	documentation	layer
55	$_c_{I,S} := F^{source}_{N,I} \star^N c_{N,S}$	interface equation	macroscopic $\rightarrow$ reactions
57	$_x_{I,S} := F^{source}_{N,I} \star^N x_{N,S}$	interface equation	macroscopic $\rightarrow$ reactions
59	$_T_I := F^{source}_{N,I} \star^N T_N$	interface equation	macroscopic $\rightarrow$ reactions
60	$T_{N,p} := (F^{sink}_{N,I} \cdot _T_I) \star^I S_{I,p}$	interface equation	reactions
143	$_mv_I_I := F^{source}_{N,I} \star^N \bar{T}_N$	interface equation	macroscopic $\rightarrow$ control